

Rethinking Spiking Self-Attention Mechanism: Implementing α -XNOR Similarity Calculation in Spiking Transformers

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Abstract

Transformers significantly raise the performance limits across various tasks, spurring research into integrating them into spiking neural networks. However, a notable performance gap remains between existing spiking Transformers and their artificial neural network counterparts. Here, we first analyze the cause of this gap and attribute it to the dot product's ineffectiveness in measuring similarity between spiking queries and keys, due to numerous non-spiking events. To address this, we propose a novel α -XNOR similarity measure tailored for spike trains. It redefines the correlation between non-spike pairs as a specific value α , effectively overcoming the limitations of dot-product similarity. Furthermore, considering the sparse nature of spike trains where spikes carry more information than non-spikes, the α -XNOR similarity correspondingly highlights the distinct importance of spikes over non-spikes. Extensive experiments demonstrate that α -XNOR similarity significantly improves performance across different spiking Transformer architectures on various static and neuromorphic datasets, further revealing the potential of spiking Transformers.

1. Introduction

Transformer in Artificial Neural Networks (ANNs) [10, 11, 31, 38] offers an effective method for processing long sequences in parallel, significantly improving performance in various tasks [6, 10, 16, 23, 37]. However, its reliance on large-scale parameters [31] and the complexity of its self-attention mechanism [52] result in high energy consumption and hardware costs [18, 41]. Consequently, developing energy-efficient and high-performance Transformer [3, 24] remains a prominent research focus.

Spiking Neural Networks (SNNs) [17, 26, 32, 33], as the third generation of neural networks, gain significant attention due to their superior energy efficiency [33, 42].

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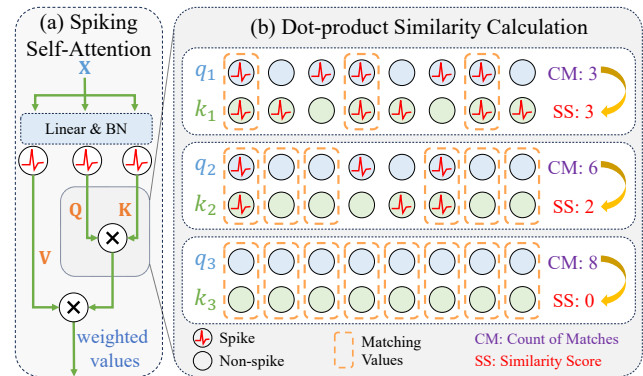


Figure 1. (a) Illustration of the Spiking Self-Attention (SSA) mechanism. (b) Similarity calculation in SSA, highlighting dot-product limitations with three spike train pairs: (1) q_1 and k_1 have 3 matching values and a consistent score of 3; (2) q_2 and k_2 have 6 matching values but only score 2; and (3) identical all-zero train q_3 and k_3 yield a clearly unreasonable score of 0.

Spiking neurons fire only when activated and remain silent in others. This event-driven mechanism [5] greatly enhances energy efficiency by promoting sparse synaptic operations and avoiding multiply-accumulate (MAC) computations [7, 34, 49]. Recently, numerous researchers focus on developing spiking Transformer models, such as Spikformer [54], Spike-driven Transformers [44, 46], and SpikingResformer [36]. These approaches enhance SNNs' performance limits in various aspects. However, a substantial performance gap remains compared to their ANNs' counterparts [40]. Through theoretical analysis, we attribute this gap to the unsuitability of dot-product similarity for the binary nature of spike trains in existing spiking Transformers.

In ANNs, self-attention calculates token similarities by taking the dot product of Queries (Q) and Keys (K), assigning similarity scores to Values (V) to model human selective attention [12]. However, the sparse, event-driven nature of SNNs results in numerous non-spikes (represented as 0) and a limited number of spikes (represented as 1) in spiking Q

and K . Due to the unique behavior of 0 in multiplication, both 0×0 and 0×1 result in 0, making effective distinction challenging in dot product. As illustrated in Fig.1, the relationship between dot-product similarity and true matching for spike trains is examined. Even when Q and K vectors are entirely identical (e.g., q_3 and k_3 in Fig.1), the presence of zero multiplications causes the dot-product similarity to diverge significantly from the true matches. Thus, designing a specialized similarity calculation method for spike trains could improve spiking Transformers' performance.

In this paper, we propose the α -XNOR similarity calculation specifically designed for SNNs. The α -XNOR similarity redefines the correlation of non-spike pairs as a specific value α to address the ineffectiveness in similarity assessments. Furthermore, given the sparse nature of spike trains, spikes convey more critical information than non-spikes. Therefore, the α -XNOR similarity method also emphasizes distinguishing the relative roles of spikes and non-spikes. Based on α -XNOR similarity, we integrate it into the spiking self-attention mechanism and develop an α -XNOR Spiking Self-Attention (α -SSA) module. Extensive experiments demonstrate that α -SSA significantly improves the performance of spiking Transformers across various architectures. The main contributions of this paper are summarized as follows:

- We thoroughly analyze the reasons why the dot product is unsuitable for calculate spiking similarity. Our findings reveal that the unique multiplicative properties of non-spike hinder the dot-product's ability to accurately capture the similarity between sparse spiking Q and K .
- To address this issue, we propose the α -XNOR similarity, which redefines the correlation of non-spike pairs as a specific value α . This adjustment not only overcomes the ineffectiveness of dot-product similarity but also emphasizes the distinct importance of spikes over non-spikes, enabling richer information representation.
- Based on the α -XNOR similarity, we propose the α -XNOR Spiking Self-Attention (α -SSA) module. Extensive experiments show that α -SSA achieves significant performance improvements across various Transformer architectures on both static and neuromorphic datasets.

2. Related Work

Spiking Neural Networks: The event-driven mechanism of SNNs greatly enhances energy efficiency, providing a critical advantage for compute-constrained edge devices. The advent of ANN-to-SNN conversion [4, 22] and direct training methods [20, 42] significantly reduce the challenges of training high-performance SNNs. Leveraging these advanced learning algorithms, some studies [25, 51] propose deep residual SNNs, while others [43, 48] develop multi-dimensional spiking attention mechanisms, achieving competitive results on ImageNet [8, 35] and various neuro-

morphic datasets [1, 28, 30]. Further research explores parallel training strategies [15] and shortcuts residual connections [13, 25] for SNNs in the ResNet framework, extending SNNs' applicability to diverse tasks. However, despite these advances, a notable performance gap remains between the deep SNN architectures and the latest Transformers.

Spiking Transformers: Recently, growing attention has been paid to high-performance, energy-efficient spiking Transformers. Some models [39, 53, 54] adopt a single-stage Vision Transformer (ViT) architecture [11], while recent approaches [36, 44, 46] favor a multi-stage architecture [31] to enable hierarchical feature extraction. For instance, Spikformer [54] first proposes spike-based self-attention computation, establishing the first spiking Transformer. Building on this, Spikingformer [53] introduces a hardware-friendly residual learning architecture that avoids non-spike computations in SNNs. Additionally, the Spike-driven Transformer [44] incorporates the Hadamard product into the self-attention module, achieving a fully spike-driven mechanism. Furthermore, SpikingResformer [36] integrates a Dual Spike self-attention module, enhancing both performance and energy efficiency. Nonetheless, these models primarily utilize self-attention as a token mixer [47], without paying attention to exploring effective similarity calculations suited to spike trains [2, 50]. Therefore, designing spiking self-attention mechanisms tailored to the binary nature of SNNs is essential for further advancement.

3. Preliminaries

3.1. Leaky Integrate-and-Fire model

The Leaky Integrate-and-Fire (LIF) model [32] is one of the most widely used spiking neuron models in SNNs due to its efficiency in mimicking biological neurons. In a LIF neuron, the membrane potential $\mathbf{H}[t]$ accumulates synaptic input and decays over time. When $\mathbf{H}[t]$ exceeds the threshold V_{th} , the neuron emits a spike. After firing, $\mathbf{H}[t]$ is reset to V_{reset} , and the postsynaptic potential $\mathbf{U}[t]$ continues to evolve, decaying back to the resting potential if no further input arrives. These dynamics are governed by:

$$\mathbf{H}[t] = \tau \mathbf{U}[t-1] + \mathbf{W}\mathbf{S}[t], \quad (1)$$

$$\mathbf{S}[t] = \Theta(\mathbf{H}[t] - V_{th}), \quad (2)$$

$$\mathbf{U}[t] = \mathbf{H}[t](1 - \mathbf{S}[t]) + V_{reset}\mathbf{S}[t], \quad (3)$$

where τ is the decay constant, $\mathbf{W}$ the synaptic weight matrix, $\mathbf{S}[t]$ the input spike at time t , and $\Theta(\cdot)$ the Heaviside function. $\Theta(\cdot)$ outputs 1 when a spike is fired, and 0 for the absence of a spike (i.e., a non-spike). For brevity, we denote the spiking neuron layer as $\mathcal{SN}(\cdot)$ in subsequent sections.

3.2. Vanilla Self-attention

The self-attention mechanism, a key component of the Transformer model, efficiently processes large-scale data by

capturing global dependencies and flexibly allocating attention across tokens in the input sequence [38]. Let the input matrix be $\mathbf{X} \in \mathbb{R}^{n \times d}$, where n is the sequence length and d is the embedding dimension. The vanilla self-attention (VSA) first projects $\mathbf{X}$ into query $\mathbf{Q}$, key $\mathbf{K}$, and value $\mathbf{V}$ matrices using three distinct weight matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ as follows:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V. \quad (4)$$

Next, the similarity between the elements is computed to determine the level of attention each token should receive. This is achieved by calculating the dot-product similarity between the corresponding tokens in $\mathbf{Q}$ and $\mathbf{K}$ and generates the attention scores:

$$\text{Score} = \text{Sim}_{\text{DP}}(\mathbf{Q}, \mathbf{K}) = \mathbf{Q}\mathbf{K}^\top, \quad (5)$$

where $\text{Sim}_{\text{DP}}(\cdot)$ denote the similarity function by dot product. Then, a scaling factor $\sqrt{d}$ is applied to prevent the dot-product values from becoming large, and $\text{Softmax}(\cdot)$ is utilized to obtain the attention scores. The final output $\mathbf{Z}$ is then computed as the weighted sum of the value matrix $\mathbf{V}$:

$$\mathbf{Z} = \text{Softmax}\left(\frac{\text{Score}}{\sqrt{d}}\right) \mathbf{V}. \quad (6)$$

Based on the dot-product similarity, the model is able to flexibly distribute attention across the sequence, allowing long-range dependencies capture.

3.3. Spiking Self-Attention

Building on VSA, Spikformer [54] introduced a spiking variant called Spiking Self-Attention (SSA). For the input $\mathbf{X} \in \mathbb{R}^{n \times d}$, the queries, keys, and values are first computed using learnable weight matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, $\mathbf{W}_V$ and then transformed into spike trains as follows:

$$\mathbf{Q} = \mathcal{SN}(\text{BN}(\mathbf{X}\mathbf{W}_Q)), \quad (7)$$

$$\mathbf{K} = \mathcal{SN}(\text{BN}(\mathbf{X}\mathbf{W}_K)), \quad (8)$$

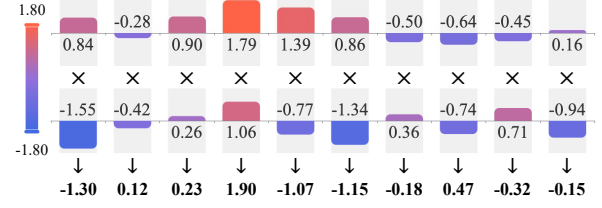
$$\mathbf{V} = \mathcal{SN}(\text{BN}(\mathbf{X}\mathbf{W}_V)), \quad (9)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{T \times n \times d}$, $\text{BN}(\cdot)$ denotes batch normalization, and $\mathcal{SN}(\cdot)$ refers to the spiking neuron layer, ensuring the attention mechanism operates entirely in spike form. Similar to VSA, SSA computes the similarity between spiking $\mathbf{Q}$ and $\mathbf{K}$ using a dot-product operation:

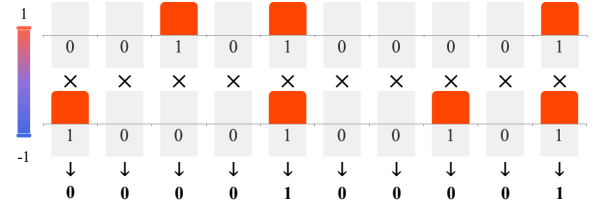
$$\text{Score} = \text{Sim}_{\text{DP}}(\mathbf{Q}, \mathbf{K}) = \mathbf{Q}\mathbf{K}^\top. \quad (10)$$

Unlike VSA, SSA omits the softmax operation while retaining scaling to control the magnitude of the attention output. The attention output Attn is computed as a weighted sum of $\mathbf{V}$, scaled by factor s , and then converted to spiking form:

$$\text{Attn}' = s \cdot \text{Score}\mathbf{V}, \quad \text{Attn} = \mathcal{SN}(\text{Attn}'). \quad (11)$$



(a) Dot-product operation in ANNs.



(b) Dot-product operation in SNNs.

Figure 2. Dot-product operation in ANNs and SNNs. (a): Dot-product operation in ANNs, with data sampled from VSA. (b): Dot-product operation in SNNs, with data sampled from SSA.

Finally, the attention result passes through a linear layer and batch normalization, followed by the spiking neuron layer, to produce the final SSA output $\mathbf{Z}$:

$$\mathbf{Z} = \mathcal{SN}(\text{BN}(\text{Linear}(\text{Attn}))), \quad (12)$$

where $\text{Linear}(\cdot)$ is the fully connected layer. Thus, SSA provides a spiking version of self-attention, enabling the spiking Transformer to capture global dependencies efficiently while maintaining scalability.

4. Methods

4.1. Spiking Dot-Product Similarity Ineffectiveness

In ANNs, the attention mechanism effectively captures relevant information by using dot products to calculate query-key similarity, where higher similarity increases the weight of the corresponding value. However, in SNNs, the spike-driven nature poses two major challenges, making the spiking dot products ineffective for calculating similarity between spike trains.

Firstly, the unique mathematical properties of 0 prevent dot product from effectively capturing correlations involving non-spike events (represented as 0) in SNNs. As shown in Fig. 1, in the first spike train pair q_1 and k_1 , the 3 matching values yield a dot-product similarity score of 3, which is consistent. However, this consistency relies on the absence of 0-0 pairs. In the second pair q_2 and k_2 , despite 6 matching values, the dot-product score is only 2, as 0-0 pairs do not contribute in multiplication. A more extreme case is the third pair q_3 and k_3 , where both spike trains are entirely identical, yet the dot-product similarity score is 0, which

is significantly unreasonable. Notably, due to the sparsity of spike trains, pairs like the second are the most common. Thus, the dot-product similarity is ineffective in SSA due to the multiplication properties of 0.

Secondly, dot-product operations in ANNs effectively capture a wide range of attentional preferences, with sign bits indicating the direction of correlation between corresponding elements. In contrast, due to the binary nature of spike trains in SNNs, dot product results are limited to two states. Additionally, they lack sign bits to denote the direction of correlation. Consequently, these limitations lead to a significant deficiency in information capacity during dot-product similarity calculations. As depicted in Fig.2, compared to the rich dot product outcomes in ANNs, the results become increasingly sparse following dot-product calculations in SNNs and fail to represent the correlation direction.

4.2. α -XNOR Similarity

To address the issue wherein matching non-spike pairs do not contribute to the dot-product similarity score, we intuitively expect that identical pairs, whether they are spikes or non-spikes, contribute equally to the similarity score. Thus, we replace the multiplication operation in dot product with an operation where matching values contribute a score of 1 while differing values contribute a score of 0. This adjustment aligns with the XNOR operation in binary logic:

$$\text{XNOR}(x, y) = \begin{cases} 1, & \text{if } x = y, \\ 0, & \text{if } x \neq y. \end{cases} \quad (13)$$

The XNOR operation therefore measures the similarity by evaluating the exact matches between two spike trains.

However, most current SNNs operate at a low firing rate, spikes typically represent rare but significant events, making each spike occurrence carry a higher informational value (details provided in Suppl. B). Consequently, spikes can be considered more important than non-spikes within the spiking data. Building on this insight, we further adapt the XNOR operation to distinguish between matching spikes and matching non-spikes. Specifically, we assign a lower weight to matching non-spikes, making them less significant than matching spikes yet more significant than non-matching pairs. This weight, denoted as α , is set between 0 and 1. The modified operation is formulated as:

$$\alpha\text{-XNOR}(x, y) = \begin{cases} 1, & \text{if } x = y = 1, \\ \alpha, & \text{if } x = y = 0, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

Here, $\alpha\text{-XNOR}(\cdot)$ denotes the modified XNOR operation.

Since $\alpha\text{-XNOR}(\cdot)$ is non-differentiable, we define custom gradients to ensure smooth gradient flow during back-propagation:

$$\frac{\partial \alpha\text{-XNOR}(x, y)}{\partial x} = -2\alpha x + (1 + \alpha)y, \quad (15)$$

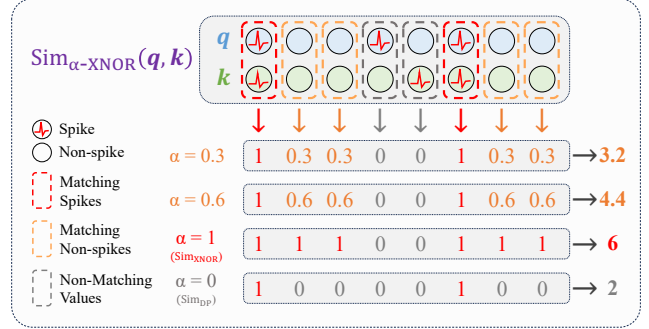


Figure 3. Illustration of α -XNOR similarity with α values of 0.3, 0.6, 1, and 0. Cases where $\alpha = 1$ and $\alpha = 0$ correspond to XNOR similarity and dot-product similarity, respectively.

$$\frac{\partial \alpha\text{-XNOR}(x, y)}{\partial y} = -2\alpha y + (1 + \alpha)x. \quad (16)$$

These gradients are derived from a differentiable function that is numerically equivalent to $\alpha\text{-XNOR}(\cdot)$:

$$\alpha\text{-XNOR}_{\text{grad}}(x, y) = (1 - \alpha)xy - \alpha(x - y)^2 + \alpha. \quad (17)$$

By adopting the gradients in Eq. 17, the α -XNOR operation achieves both computational efficiency and a smooth gradient flow, facilitating effective parameter updates during training.

Building on $\alpha\text{-XNOR}(\cdot)$, we define the α -XNOR similarity between two spike trains $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ as:

$$\text{Sim}_{\alpha\text{-XNOR}}(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^d \alpha\text{-XNOR}(x_i, y_i). \quad (18)$$

For notational simplicity, we overload $\text{Sim}_{\alpha\text{-XNOR}}(\cdot)$ to also support batched similarity calculation between spike matrices. Given $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n \times d}$, we define $\text{Sim}_{\alpha\text{-XNOR}}(\mathbf{X}, \mathbf{Y}) \in \mathbb{R}^{n \times n}$ as the pairwise α -XNOR similarity matrix between all tokens in $\mathbf{X}$ and $\mathbf{Y}$:

$$[\text{Sim}_{\alpha\text{-XNOR}}(\mathbf{X}, \mathbf{Y})]_{ij} = \text{Sim}_{\alpha\text{-XNOR}}(\mathbf{X}_i, \mathbf{Y}_j), \quad (19)$$

where i, j index the tokens and range from 1 to n .

Additionally, there are two special cases for the α -XNOR similarity: when $\alpha = 0$, $\alpha\text{-XNOR}(\cdot)$ reduces to the traditional dot-product similarity (Sim_{DP}), where only matching spikes contribute; and when $\alpha = 1$, it becomes the original XNOR operation, referred to as XNOR similarity (Sim_{XNOR}), where all matching values contribute equally, regardless of being spikes or non-spikes.

As illustrated in Fig. 3, α -XNOR similarity effectively addresses the issue of non-spike pairs making no contribution in dot-product similarity. Compared to XNOR similarity, α -Similarity further differentiates the contributions of spikes and non-spikes by assigning less weight to non-spike matches, resulting in a more balanced and accurate similarity score for spiking data.

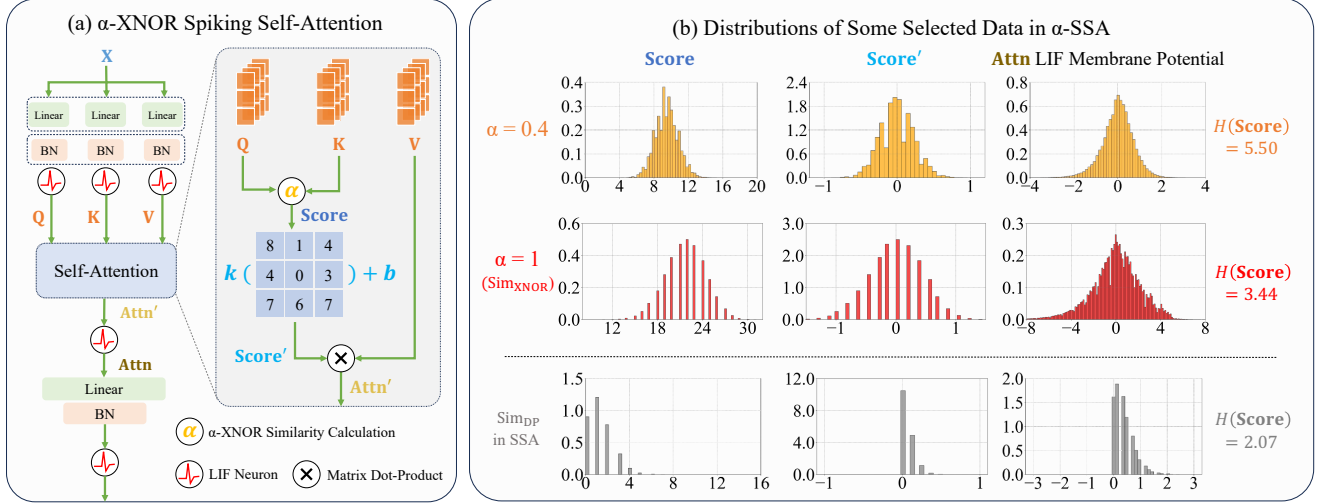


Figure 4. α -XNOR Spiking Self-Attention (α -SSA) mechanism. (a) Overall architecture of α -SSA. (b) Distributions of similarity scores calculated using α -XNOR similarity with $\alpha = 0.4$ and $\alpha = 1$ (XNOR similarity) before and after linear transformation, alongside the membrane potential distribution prior to firing in the LIF neuron that generates the self-attention outputs $Attn$. For a clear comparison, corresponding distributions from the original SSA are also shown.

Moreover, α -XNOR similarity also significantly increases the information representation within the spiking self-attention module. In the following, we present a theoretical analysis for this. We begin with the information entropy as our criterion, which is defined as:

$$H(X) = - \sum_{i=1}^n p_i \log_2 p_i, \quad (20)$$

where X is a random variable with outcome probabilities p . Higher entropy values signify greater variability and informational content in the similarity scores. The entropy can be calculated based on the distribution of the scores.

Theorem 1. Let $q, k \in \{0, 1\}^d$ be independent spike trains with firing rates f_q and f_k , respectively. The similarity scores for dot-product and XNOR similarity both follow a binomial distribution:

$$\text{Sim}_{DP}(q, k) \sim \text{Bin}(d, p_1), \quad (21)$$

$$\text{Sim}_{XNOR}(q, k) \sim \text{Bin}(d, p_1 + p_2), \quad (22)$$

where $p_1 = f_q f_k$, and $p_2 = (1 - f_q)(1 - f_k)$, represent the probability of matching spikes and non-spikes, respectively.

Theorem 1 (proof in Suppl. A.1) shows that scores generated by the two special cases of α -XNOR similarity, dot-product similarity and XNOR similarity, both approximately follow a binomial distribution, with different probability parameters. However, the score distribution of α -XNOR similarity in the general case is much more complex.

Theorem 2. Let $q, k \in \{0, 1\}^d$ be independent spike trains with firing rates f_q and f_k , respectively. The score calculated by α -XNOR similarity is distributed as:

$$P(s = k + l\alpha) = \frac{d!}{k! l! (d - k - l)!} \cdot p_1^k p_2^l p_3^{d-k-l}, \quad (23)$$

for $0 \leq k \leq d$ and $0 \leq l \leq d - k$, where $p_1 = f_q f_k$, $p_2 = (1 - f_q)(1 - f_k)$, and $p_3 = 1 - p_1 - p_2$.

Theorem 2 (proof in Suppl. A.2) demonstrates that the similarity score calculated by α -XNOR similarity approximately follows a weighted multinomial distribution, where matching spikes and non-spikes contribute to the score with weights 1 and α , respectively. The introduction of α increases the complexity of the score distribution. By distinguishing between spikes and non-spikes, α -XNOR similarity captures a broader range of variability in similarity scores, resulting in higher information entropy. The following Theorem 3 further reveals the relationships among scores produced by these similarity calculations.

Theorem 3. Let $q, k \in \{0, 1\}^d$ be independent spike trains with firing rates $f_q, f_k \in (0, 1)$ and $\alpha \in (0, 1)$. Then the entropies satisfy: $H(\text{Sim}_{\alpha-XNOR}(q, k)) > H(\text{Sim}_{DP}(q, k))$ and $H(\text{Sim}_{\alpha-XNOR}(q, k)) > H(\text{Sim}_{XNOR}(q, k))$.

Theorem 3 (proof provided in Suppl. A.3) shows that the α -XNOR similarity achieves higher information entropy compared to dot-product and XNOR similarities, reflecting its enhanced capacity for information representation. Fig. 4(b) compares the similarity scores calculated by these methods under the parameters $f_q = f_k = 0.2$, $d = 32$, and

$\alpha = 0.4$. Unlike dot-product and XNOR similarities, which generate highly discretized distributions with limited variability, the α -XNOR similarity produces smoother, more centrally distributed scores. This broader value range enables richer information representation and more nuanced similarity assessments. The information entropy of scores from α -XNOR similarity is 5.50, significantly exceeding the 2.07 and 3.44 from dot-product and XNOR similarities, respectively, highlighting the effectiveness of α -XNOR similarity for spiking data.

4.3. α -XNOR Spiking Self-Attention

In Sec. 4.2, we introduce α -XNOR similarity to address dot-product limitations, and differentiate spikes and non-spikes in similarity calculations. Leveraging this spiking similarity calculation, we develop an improved spiking self-attention module, termed as α -XNOR Spiking Self-Attention (α -SSA).

The overall architecture of α -SSA is shown in Fig. 4 (a). Initially, the spiking $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ matrices are generated similarly to SSA using Eq. 4. Then, we calculate the similarity scores with α -XNOR similarity instead of the original dot-product similarity:

$$\mathbf{Score} = \text{Sim}_{\alpha\text{-XNOR}}(\mathbf{Q}, \mathbf{K}). \quad (24)$$

Following this, the obtained attention scores undergo transformation. In the original SSA, only a scaling factor adjusts the score magnitudes, which is inadequate for scores based on α -XNOR similarity, as they exhibit different distribution patterns (see Fig. 4 (b)). In α -SSA, we use a linear transformation for the score with an added bias term:

$$\mathbf{Attn}' = (k \cdot \mathbf{Score} + b)\mathbf{V}, \quad (25)$$

where k and b are constants that help adjust the attention output distribution. For $\mathbf{Score}$ derived from α -XNOR, the corresponding $\mathbf{Attn}'$ has an expected value $\mathbb{E}[\mathbf{Attn}']$ and variance $\mathbb{V}[\mathbf{Attn}']$, which can be given by (proof provided in Suppl. A.4):

$$\mathbb{E}[\mathbf{Attn}'] = n f_{\mathbf{V}} (k d \gamma_1 + b), \quad (26)$$

$$\mathbb{V}[\mathbf{Attn}'] = n f_{\mathbf{V}} (k^2 d (\gamma_2 - \gamma_1^2) + (1 - f_{\mathbf{V}}) (k d \gamma_1 + b)^2), \quad (27)$$

where γ_1 and γ_2 are determined by the firing rates of $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$, denoted as $f_{\mathbf{Q}}$, $f_{\mathbf{K}}$, and $f_{\mathbf{V}}$, respectively:

$$\gamma_1 = f_{\mathbf{Q}} f_{\mathbf{K}} + \alpha (1 - f_{\mathbf{Q}}) (1 - f_{\mathbf{K}}), \quad (28)$$

$$\gamma_2 = f_{\mathbf{Q}} f_{\mathbf{K}} + \alpha^2 (1 - f_{\mathbf{Q}}) (1 - f_{\mathbf{K}}), \quad (29)$$

To achieve a balanced membrane potential distribution, we set $\mathbb{E}[\mathbf{Attn}'] = 0$ and $\mathbb{V}[\mathbf{Attn}'] = 1$, leading to the following values for k and b :

$$k = \frac{1}{\sqrt{n d (\gamma_2 - \gamma_1^2) f_{\mathbf{V}}}}, \quad b = -\frac{\gamma_1 \sqrt{d}}{\sqrt{n (\gamma_2 - \gamma_1^2) f_{\mathbf{V}}}}. \quad (30)$$

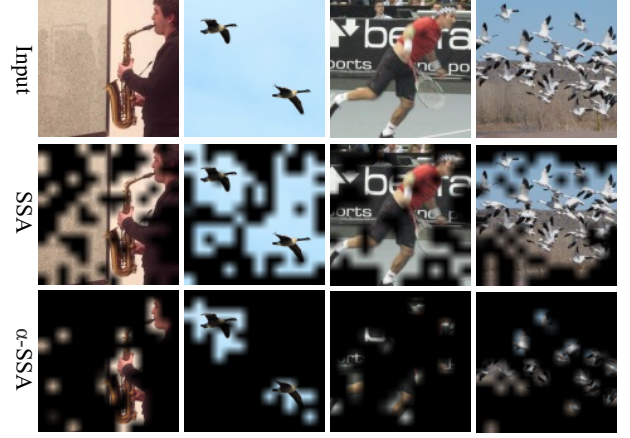


Figure 5. Attention heatmaps of α -SSA and SSA models on ImageNet1K, showing α -SSA's sparser focus on key visual features.

Here, we select the positive solution of k to ensure that the scores after the linear transformation are positively correlated with the original values.

Finally, the attention output is converted to spiking form as $\mathbf{Attn} = \mathcal{SN}(\mathbf{Attn}')$. Following [45], the multiplication of floating-point data in Eq. 25 can be merged into the threshold of this spiking neuron layer, preserving the spike-driven nature of SNNs. Subsequently, $\mathbf{Attn}$ is processed using Eq. 12 in SSA to generate the final output.

Fig. 4 (b) shows the data distributions in α -SSA alongside those in SSA for comparison. In α -SSA, scores calculated using α -XNOR yield not only more reasonable similarity values but also a distribution closer to normal with significantly higher information entropy. By comparison, the dot-product similarity in SSA produces a positively skewed distribution with lower entropy. Furthermore, the simple scaling in SSA results in a discrete and skewed membrane potential distribution, which previous studies [19, 21] suggest may impact model performance. Meanwhile, the linear transformation in α -SSA with XNOR similarity and α -XNOR produces a more balanced membrane potential distribution. These improvements in similarity scores, distribution balance, and entropy in α -SSA highlight its advantage in capturing richer information from spiking input patterns for spiking self-attention mechanisms.

5. Experiments

5.1. Image Classification

In image classification experiments, we evaluate the proposed α -SSA module on both static and dynamic image datasets, including ImageNet [8], CIFAR-10, CIFAR-100 [27], and CIFAR10-DVS [28]. We implement α -SSA module within both Single-stage ViT (S-ViT) and Multi-stage ViT (M-ViT) architectures and conduct detailed com-

Table 1. Detailed comparison with other similar methods on ImageNet

Type	Method	Architecture	Time Step	Param. (M)	Acc. (%)
ResNet	SEW ResNet [13]	Sew ResNet-50	4	25.6	67.78
		Sew ResNet-101	4	44.6	68.76
	MS ResNet [25]	MS-ResNet-50	4	21.8	69.40
		MS-ResNet-104	4	77.3	75.30
	Att-MS-ResNet [43]	Att-ResNet-50	1	22.1	69.20
		Att-ResNet-104	4	78.4	76.30
Single-stage ViTs	Spikformer [54]	Spikformer-8-384	4	16.8	70.24
		Spikformer-8-512	4	29.7	73.38
	Spikingformer [53]	Spikingformer-8-384	4	16.8	70.45
		Spikingformer-8-512	4	29.7	74.49
	S-ViT + α-SSA ($\alpha = 0.3$)	α -SSA-ViT-8-384	4	16.8	71.36
	S-ViT + α-SSA ($\alpha = 0.3$)	α -SSA-ViT-8-512	4	29.7	75.39
Multi-stage ViTs	Spike-driven Transformer [44]	Spike-driven-8-384	4	16.8	72.28
		Spike-driven-8-512	4	29.7	74.57
	SpikingResformer [36]	SpikingResformer-S	4	17.8	75.95
		SpikingResformer-M	4	35.5	77.24
	Meta-SpikeFormer [45]	Meta-SpikeFormer-384	4	15.1	74.10
		Meta-SpikeFormer-512	4	55.4	79.70
	M-ViT + α-SSA ($\alpha = 0.3$)	α -SSA-Swin-10-384	4	31.8	77.89
	M-ViT + α-SSA ($\alpha = 0.3$)	α -SSA-Swin-10-512	4	54.6	80.62

parisons with the latest spiking Transformers [36, 44, 45, 53, 54]. The results are as follows:

ImageNet: For the ImageNet dataset, we set the input image dimensions to 224×224 , with a batch size of 128 and a total of 300 training epochs. In this set of experiments, the value of α is set to 0.3 for all α -SSA layers across different models. As shown in Table 1, our α -SSA module enhances performance across both S-ViT and M-ViT architectures. Specifically, in the S-ViT architecture, α -SSA increases accuracy to 75.39% in the 512-dimensional setting. Similarly, in the M-ViT Transformer architecture, α -SSA achieves 80.62% in the 512-dimensional setting. Additionally, we compared the attention maps of the original SSA and our α -SSA. As shown in Fig. 5, α -SSA demonstrates sparser attention with a sharper focus on key visual features.

CIFAR10/100 and CIFAR10-DVS: CIFAR10 and CIFAR100 [27] are widely used benchmarks for image classification, while CIFAR10-DVS [28] is an event-based dataset specifically designed for SNN models. In both architectures, we conducted experiments with two different values of $\alpha = 0.3$ and $\alpha = 0.5$. As demonstrated in Table 2, our proposed α -SSA module consistently boosts performance across various architectures and datasets while maintaining model simplicity. Notably, in M-ViT mod-

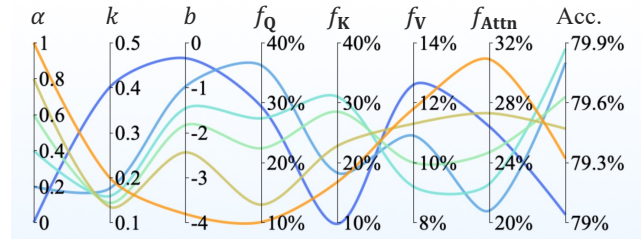


Figure 6. Parallel coordinate plots of α -SSA parameters (α , k , b) and metrics (firing rates for $\mathbf{Q}$ (f_Q), $\mathbf{K}$ (f_K), $\mathbf{V}$ (f_V), attention output (f_{Attn}), and accuracy), showing correlations between α and other variables. Results are based on CIFAR100 with α -SSA in the Spikformer architecture.

els, α -SSA achieves exceptional outcomes, attaining accuracies of 96.15% on CIFAR10, 81.08% on CIFAR100, and 84.24% on CIFAR10-DVS, underscoring its effectiveness in enhancing model robustness and accuracy. Notably, The performance improvements in the M-ViT architecture are less significant compared to those in the S-ViT architecture. This can be attributed to the relatively minor role of self-attention modules in the Swin Transformer’s design, by findings in metaformer [47] research.

Table 2. Detailed comparison with other similar methods across CIFAR10, CIFAR100 and CIFAR10-DVS

Type	Method	CIFAR10		CIFAR100		CIFAR10-DVS	
		Param. (M)	Acc. (%)	Param. (M)	Acc. (%)	Param. (M)	Acc. (%)
ResNet	PLIF [14]	-	93.50	-	-	-	74.8
	tdBN [51]	-	93.2	-	-	-	67.8
	DSpike [29]	-	94.25	-	74.24	-	75.4
	TET [9]	12.63	94.44	12.63	74.47	-	77.33
Single-stage ViTs	Spikformer [54]	9.32	95.51	9.32	78.21	2.57	80.9
	Spikingformer [53]	9.32	95.81	9.32	79.09	2.57	81.3
	S-ViT + α-SSA ($\alpha = 0.5$)	9.32	95.92	9.32	79.97	2.57	81.45
	S-ViT + α-SSA ($\alpha = 0.3$)	9.32	95.97	9.32	80.22	2.57	81.64
Multi-stage ViTs	Spike-driven Transformer [44]	10.28	95.60	10.28	78.40	2.57	80.0
	M-ViT + α-SSA ($\alpha = 0.5$)	6.50	96.05	6.50	80.66	1.54	82.24
	M-ViT + α-SSA ($\alpha = 0.3$)	6.50	96.15	6.50	81.08	1.54	82.22

5.2. Study of Different α

In the α -SSA module, α serves as a critical parameter. We conducted experiments on the Spikformer architecture on CIFAR100 dataset to investigate the effect of varying α values on metrics. These metrics include the linear scaling factors k and b ; the spiking firing rates f_Q , f_K , f_V , and f_{Attn} responsible for generating $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$, and the spiking self-attention output Attn , respectively; and the network’s performance accuracy (Acc.). We set α to $[0, 0.2, 0.4, 0.6, 0.8, 1]$, where $\alpha = 0$ and $\alpha = 1$ correspond to dot-product similarity and XNOR similarity, respectively, while intermediate values represent α -XNOR similarities. Fig. 6 illustrates the average values of these metrics, and detailed data across different α -SSA layers are provided in Suppl.D. The results show a clear negative correlation between α and the parameters k and b . The parameters f_Q , f_K , f_V , and f_{Attn} exhibit no clear correlation and present a complex interaction, with f_V showing relatively minimal variation. Moreover, model accuracy peaks when α ranges from 0.2 to 0.4, indicating that this range optimally reflects the differential importance of spikes and non-spikes in attention scoring.

5.3. Ablation Study

To evaluate the effectiveness of our proposed model’s components, we conducted an ablation study on the CIFAR100 dataset using Spikformer as the baseline. We replaced Spikformer’s SSA with our proposed α -SSA, testing three values of α : 0, 1, and 0.3. For $\alpha = 0$, the original SSA replaces only the linear transformations of attention scores, maintaining the traditional dot-product similarity. For $\alpha = 1$ and $\alpha = 0.3$, the model employs XNOR similarity and α -XNOR similarity, respectively. As shown in Table 3, adding the linear transformations for attention scores increased accuracy from 78.21% to 79.04%. Incorporating

Table 3. Ablation study on CIFAR100 dataset

Model	Param. (M)	Acc. (%)
Spikformer (baseline)	9.32	78.21
+ α -SSA ($\alpha = 0$)	9.32	79.04
+ α -SSA ($\alpha = 1$)	9.32	79.33
+ α -SSA ($\alpha = 0.3$)	9.32	80.22

XNOR similarity resulted in a further 0.29% gain. Finally, the α -SSA with $\alpha = 0.3$ achieved an accuracy of 80.22%, marking a 2.01% improvement over the baseline. These results demonstrate that both the introduction of α -SSA and the linear transformations of attention scores yield substantial gains with minimal increase in parameters.

6. Conclusion

This work identifies a fundamental challenge in spiking Transformers: the dot product is ineffective for calculating spiking similarity due to the binary and sparse nature of spike trains. To address this issue, we propose a novel α -XNOR similarity method specifically tailored to spike trains, which effectively resolves the limitations of dot-product similarity caused by numerous non-spiking events. Moreover, given the sparse nature of spike trains, where spikes carry more information than non-spikes, the α -XNOR similarity emphasizes the distinct importance of spikes, with enhanced representation capability in spiking self-attention. Extensive experiments demonstrate significant performance improvements across various Transformer architectures. These advancements bring us closer to realizing the full potential of spiking Transformers in efficient, neuromorphic computing applications.

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References

- [1] Arnon Amir, Brian Taba, David Berg, Timothy Melano, Jeffrey McKinstry, Carmelo Di Nolfo, Tapan Nayak, Alexander Andreopoulos, Guillaume Garreau, Marcela Mendoza, et al. A low power, fully event-based gesture recognition system. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 7243–7252, 2017. 2
- [2] Sander M Bohte, Joost N Kok, and Johannes A La Poutré. Spikeprop: backpropagation for networks of spiking neurons. In *ESANN*, pages 419–424. Bruges, 2000. 2
- [3] Han Cai, Chuang Gan, Tianzhe Wang, Zhekai Zhang, and Song Han. Once-for-all: Train one network and specialize it for efficient deployment. *arXiv preprint arXiv:1908.09791*, 2019. 1
- [4] Yongqiang Cao, Yang Chen, and Deepak Khosla. Spiking deep convolutional neural networks for energy-efficient object recognition. *International Journal of Computer Vision*, 113:54–66, 2015. 2
- [5] Stefano Caviglia, Maurizio Valle, and Chiara Bartolozzi. Asynchronous, event-driven readout of posfet devices for tactile sensing. In *2014 IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 2648–2651. IEEE, 2014. doi: [10.1109/iscas.2014.6865717](https://doi.org/10.1109/iscas.2014.6865717). 1
- [6] Chun-Fu Richard Chen, Quanfu Fan, and Rameswar Panda. Crossvit: Cross-attention multi-scale vision transformer for image classification. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 357–366, 2021. 1
- [7] Mike Davies, Narayan Srinivasa, Tsung-Han Lin, Gautham Chinya, Yongqiang Cao, Sri Harsha Choday, Georgios Dimou, Prasad Joshi, Nabil Imam, Shweta Jain, et al. Loihi: A neuromorphic manycore processor with on-chip learning. *Ieee Micro*, 38(1):82–99, 2018. 1
- [8] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009. 2, 6
- [9] Shikuang Deng, Yuhang Li, Shanghang Zhang, and Shi Gu. Temporal efficient training of spiking neural network via gradient re-weighting. *arXiv preprint arXiv:2202.11946*, 2022. 8
- [10] Jacob Devlin. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018. 1
- [11] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 1, 2
- [12] Oliver Eberle, Stephanie Brandl, Jonas Pilot, and Anders Søgaard. Do transformer models show similar attention patterns to task-specific human gaze? In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4295–4309, 2022. 1
- [13] Wei Fang, Zhaofei Yu, Yanqi Chen, Tiejun Huang, Timothée Masquelier, and Yonghong Tian. Deep residual learning in spiking neural networks. *Advances in Neural Information Processing Systems*, 34:21056–21069, 2021. 2, 7
- [14] Wei Fang, Zhaofei Yu, Yanqi Chen, Timothée Masquelier, Tiejun Huang, and Yonghong Tian. Incorporating learnable membrane time constant to enhance learning of spiking neural networks. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 2661–2671, 2021. 8
- [15] Wei Fang, Zhaofei Yu, Zhaokun Zhou, Ding Chen, Yanqi Chen, Zhengyu Ma, Timothée Masquelier, and Yonghong Tian. Parallel spiking neurons with high efficiency and ability to learn long-term dependencies. *Advances in Neural Information Processing Systems*, 36, 2024. 2
- [16] Yuxin Fang, Bencheng Liao, Xinggang Wang, Jiemin Fang, Jiyang Qi, Rui Wu, Jianwei Niu, and Wenyu Liu. You only look at one sequence: Rethinking transformer in vision through object detection. *Advances in Neural Information Processing Systems*, 34:26183–26197, 2021. 1
- [17] Wulfram Gerstner and Werner M Kistler. *Spiking neuron models: Single neurons, populations, plasticity*. Cambridge university press, 2002. 1
- [18] Benjamin Graham, Alaaeldin El-Nouby, Hugo Touvron, Pierre Stock, Armand Joulin, Hervé Jégou, and Matthijs Douze. Levit: a vision transformer in convnet’s clothing for faster inference. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 12259–12269, 2021. 1
- [19] Yufei Guo, Xinyi Tong, Yuanpei Chen, Liwen Zhang, Xiaode Liu, Zhe Ma, and Xuhui Huang. Rectdis-snn: Rectifying membrane potential distribution for directly training spiking neural networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 326–335, 2022. 6
- [20] Yufei Guo, Xuhui Huang, and Zhe Ma. Direct learning-based deep spiking neural networks: a review. *Frontiers in Neuroscience*, 17:1209795, 2023. 2
- [21] Yufei Guo, Xiaode Liu, Yuanpei Chen, Liwen Zhang, Weihang Peng, Yuhang Zhang, Xuhui Huang, and Zhe Ma. Rmp-loss: Regularizing membrane potential distribution for spiking neural networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 17391–17401, 2023. 6
- [22] Bing Han, Gopalakrishnan Srinivasan, and Kaushik Roy. Rmp-snn: Residual membrane potential neuron for enabling deeper high-accuracy and low-latency spiking neural network. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 13558–13567, 2020. 2
- [23] Dongchen Han, Xuran Pan, Yizeng Han, Shiji Song, and Gao Huang. Flatten transformer: Vision transformer using

- focused linear attention. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 5961–5971, 2023. 1
- [24] Kai Han, Yunhe Wang, Qi Tian, Jianyuan Guo, Chunjing Xu, and Chang Xu. Ghostnet: More features from cheap operations. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1580–1589, 2020. 1
- [25] Yifan Hu, Lei Deng, Yujie Wu, Man Yao, and Guoqi Li. Advancing spiking neural networks toward deep residual learning. *IEEE Transactions on Neural Networks and Learning Systems*, 2024. 2, 7
- [26] Eugene M Izhikevich. Simple model of spiking neurons. *IEEE Transactions on neural networks*, 14(6):1569–1572, 2003. 1
- [27] Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. 2009. 6, 7
- [28] Hongmin Li, Hanchao Liu, Xiangyang Ji, Guoqi Li, and Luping Shi. Cifar10-dvs: an event-stream dataset for object classification. *Frontiers in neuroscience*, 11:309, 2017. 2, 6, 7
- [29] Yuhang Li, Yufei Guo, Shanghang Zhang, Shikuang Deng, Yongqing Hai, and Shi Gu. Differentiable spike: Rethinking gradient-descent for training spiking neural networks. *Advances in Neural Information Processing Systems*, 34: 23426–23439, 2021. 8
- [30] Yu Liang, Wenjie Wei, Ammar Belatreche, Honglin Cao, Zijian Zhou, Shuai Wang, Malu Zhang, and Yang Yang. Towards accurate binary spiking neural networks: Learning with adaptive gradient modulation mechanism. *arXiv preprint arXiv:2502.14344*, 2025. 2
- [31] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10012–10022, 2021. 1, 2
- [32] Wolfgang Maass. Networks of spiking neurons: the third generation of neural network models. *Neural networks*, 10(9):1659–1671, 1997. 1, 2
- [33] Timothée Masquelier, Rudy Guyonneau, and Simon J Thorpe. Spike timing dependent plasticity finds the start of repeating patterns in continuous spike trains. *PloS one*, 3(1): e1377, 2008. 1
- [34] Paul A Merolla, John V Arthur, Rodrigo Alvarez-Icaza, Andrew S Cassidy, Jun Sawada, Filipp Akopyan, Bryan L Jackson, Nabil Imam, Chen Guo, Yutaka Nakamura, et al. A million spiking-neuron integrated circuit with a scalable communication network and interface. *Science*, 345(6197):668–673, 2014. 1
- [35] Yimeng Shan, Malu Zhang, Rui-jie Zhu, Xuerui Qiu, Jason K Eshraghian, and Haicheng Qu. Advancing spiking neural networks towards multiscale spatiotemporal interaction learning. *arXiv preprint arXiv:2405.13672*, 2024. 2
- [36] Xinyu Shi, Zecheng Hao, and Zhaofei Yu. Spikingresnet: Bridging resnet and vision transformer in spiking neural networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5610–5619, 2024. 1, 2, 7
- [37] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *International conference on machine learning*, pages 10347–10357. PMLR, 2021. 1
- [38] A Vaswani. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017. 1, 3
- [39] Qingyu Wang, Tielin Zhang, Minglun Han, Yi Wang, Duzhen Zhang, and Bo Xu. Complex dynamic neurons improved spiking transformer network for efficient automatic speech recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 102–109, 2023. 2
- [40] Shuai Wang, Malu Zhang, Dehao Zhang, Ammar Belatreche, Yichen Xiao, Yu Liang, Yimeng Shan, Qian Sun, Enqi Zhang, and Yang Yang. Spiking vision transformer with saccadic attention. In *The Thirteenth International Conference on Learning Representations*, 2025. 1
- [41] Kan Wu, Jinnian Zhang, Houwen Peng, Mengchen Liu, Bin Xiao, Jianlong Fu, and Lu Yuan. Tinyvit: Fast pretraining distillation for small vision transformers. In *European conference on computer vision*, pages 68–85. Springer, 2022. 1
- [42] Yujie Wu, Lei Deng, Guoqi Li, Jun Zhu, and Luping Shi. Spatio-temporal backpropagation for training high-performance spiking neural networks. *Frontiers in neuroscience*, 12:331, 2018. 1, 2
- [43] Man Yao, Guangshe Zhao, Hengyu Zhang, Yifan Hu, Lei Deng, Yonghong Tian, Bo Xu, and Guoqi Li. Attention spiking neural networks. *IEEE transactions on pattern analysis and machine intelligence*, 45(8):9393–9410, 2023. 2, 7
- [44] Man Yao, Jiakui Hu, Tianxiang Hu, Yifan Xu, Zhaokun Zhou, Yonghong Tian, Bo Xu, and Guoqi Li. Spike-driven transformer v2: Meta spiking neural network architecture inspiring the design of next-generation neuromorphic chips. *arXiv preprint arXiv:2404.03663*, 2024. 1, 2, 7, 8
- [45] Man Yao, Jiakui Hu, Tianxiang Hu, Yifan Xu, Zhaokun Zhou, Yonghong Tian, Bo XU, and Guoqi Li. Spike-driven transformer v2: Meta spiking neural network architecture inspiring the design of next-generation neuromorphic chips. In *The Twelfth International Conference on Learning Representations*, 2024. 6, 7
- [46] Man Yao, Jiakui Hu, Tianxiang Hu, Yifan Xu, Zhaokun Zhou, Yonghong Tian, Bo Xu, and Guoqi Li. Spike-driven transformer v2: Meta spiking neural network architecture inspiring the design of next-generation neuromorphic chips. *arXiv preprint arXiv:2404.03663*, 2024. 1, 2
- [47] Weihao Yu, Mi Luo, Pan Zhou, Chenyang Si, Yichen Zhou, Xinchao Wang, Jiashi Feng, and Shuicheng Yan. Metaformer is actually what you need for vision. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10819–10829, 2022. 2, 7
- [48] Dehao Zhang, Shuai Wang, Ammar Belatreche, Wenjie Wei, Yichen Xiao, Haorui Zheng, Zijian Zhou, Malu Zhang, and Yang Yang. Spike-based neuromorphic model for sound source localization. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. 2
- [49] Jilin Zhang, Dexuan Huo, Jian Zhang, Chunqi Qian, Qi Liu, Liyang Pan, Zhihua Wang, Ning Qiao, Kea-Tiong Tang,

- and Hong Chen. 22.6 anp-i: A 28nm 1.5 pj/sop asynchronous spiking neural network processor enabling sub-o. 1 μ j/sample on-chip learning for edge-ai applications. In *2023 IEEE International Solid-State Circuits Conference (ISSCC)*, pages 21–23. IEEE, 2023. [1](#)
- [50] Malu Zhang, Jiadong Wang, Jibin Wu, Ammar Belatreche, Burin Amornpaisannon, Zhixuan Zhang, Venkata Pavan Kumar Miriyala, Hong Qu, Yansong Chua, Trevor E Carlson, et al. Rectified linear postsynaptic potential function for backpropagation in deep spiking neural networks. *IEEE transactions on neural networks and learning systems*, 33(5):1947–1958, 2021. [2](#)
- [51] Hanle Zheng, Yujie Wu, Lei Deng, Yifan Hu, and Guoqi Li. Going deeper with directly-trained larger spiking neural networks. In *Proceedings of the AAAI conference on artificial intelligence*, pages 11062–11070, 2021. [2](#), [8](#)
- [52] Lin Zheng, Chong Wang, and Lingpeng Kong. Linear complexity randomized self-attention mechanism. In *International conference on machine learning*, pages 27011–27041. PMLR, 2022. [1](#)
- [53] Chenlin Zhou, Liutao Yu, Zhaokun Zhou, Zhengyu Ma, Han Zhang, Huihui Zhou, and Yonghong Tian. Spikingformer: Spike-driven residual learning for transformer-based spiking neural network. *arXiv preprint arXiv:2304.11954*, 2023. [2](#), [7](#), [8](#)
- [54] Zhaokun Zhou, Yuesheng Zhu, Chao He, Yaowei Wang, YAN Shuicheng, Yonghong Tian, and Li Yuan. Spikformer: When spiking neural network meets transformer. In *The Eleventh International Conference on Learning Representations*, 2023. [1](#), [2](#), [3](#), [7](#), [8](#)